The Singleton Attractor: Formal Theory and Simulation of Dominant Intelligence Emergence

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https://github.com/ninjahawk/singleton-attractor

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Abstract

We formalize and test the claim that competitive environments with recursive self-improvement converge to a *singleton attractor*: one agent achieves unbounded capability advantage over all others in finite time. The model combines Yudkowsky's intelligence explosion equation ($dS/dt = S^{1-\beta}$), Omohundro's instrumental resource acquisition, and Lotka–Volterra competitive exclusion into a unified framework with five explicit assumptions. We prove three results formally and state one simulation-supported conjecture: (1) ratio divergence under resource competition (Theorem 3.2), (2) finite-time separation at the β-threshold under adversarial resource dynamics (Theorem 3.4), (3) local instability of the equal-resource equilibrium (Proposition 3.9), and (4) N-agent generalization by induction (Theorem 3.10, simultaneous-dynamics gap acknowledged). We additionally derive the exact transcendental condition for coalition external suppression with numerical solution $\alpha^* \approx 0.64$ (Theorem 6.2), and a complete stochastic blowup proof via Feller's explosion criterion (Theorem 7.1). All results are conditional on Assumption A4 (reachable β-threshold); whether A4 holds in any real system is not addressed. Eleven simulation scripts and 25 model-internal findings verify the numerical behavior of the ODE system. Key results: the β-threshold produces 12.4×10^6 times greater separation than flat dynamics; stochastic noise randomizes which agent wins but never prevents singleton formation (zero failures in 300 trials per noise level); cooperation under any structure produces zero formation failures with indistinguishable timing. We identify five conditions under which the theorem fails and characterize each quantitatively. One finding revises Bostrom's niche partitioning failure mode: separate resource niches do not prevent singleton emergence when agents share the same β-function; genuine failure requires structurally different growth ceilings.

1 Introduction

Bostrom (2005) defined the singleton as "a world order in which there is a single decision-making agency at the highest level." He argued a singleton was plausible and analyzed its properties but did not formally prove its emergence from competitive dynamics. Omohundro (2008) identified resource acquisition as a convergent instrumental goal for capable optimization processes. Yudkowsky (2013) formalized the intelligence explosion as $dS/dt = S^{1-\beta}$, showing that $\beta < 0$ produces finite-time singularity. The classical Lotka–Volterra competition model (Lotka, 1925; Volterra, 1926) establishes that two agents sharing a resource pool with even a marginal growth-rate advantage diverge in ratio as $e^{(r_1-r_2)t}$.

No prior work combines these into a unified formal model with falsifiable quantitative predictions. Good's (1965) intelligence explosion concept is qualitative. Bostrom's singleton hypothesis is historical-extrapolative. Omohundro's drives argument is mechanistic but not formalized as a dynamical system. Yudkowsky's growth equation treats single-agent dynamics. Lotka–Volterra treats biological populations without recursive self-improvement.

The central conditional of this paper is Assumption A4: that recursive self-improvement can cross from diminishing to accelerating returns—i.e., that $\beta(S)$ becomes negative for some reachable $S = T$. All results are conditional on A4. Whether A4 holds in any real system is an empirical question this paper does not answer; it is the key uncertainty between the formal results and any real-world application. Readers should note that the core argument has a near-tautological structure: A4 asserts that superexponential growth is reachable, and the theorems prove that if superexponential growth is reached, the agent dominates. The formal machinery establishes *how* this dominance occurs (finite-time, under adversarial resource dynamics, robust to noise and cooperation) but does not resolve the contested empirical premise.

This paper makes the following contributions.

(1) **A formal model** combining recursive self-improvement, instrumental resource acquisition, and competitive exclusion with five explicit assumptions (A1–A5, Section 2).

(2) **Three formal proofs and one simulation-supported conjecture** (Section 3): ratio divergence under competition (Theorem 3.2); finite-time separation at the β-threshold under adversarial resource dynamics (Theorem 3.4); local instability of the equal-resource state (Proposition 3.9, local only); N-agent singleton by induction (Theorem 3.10, simultaneous-dynamics gap acknowledged).

(3) **Two supplementary derivations** (Sections 6, 7): the exact transcendental condition for coalition external suppression (Theorem 6.2) with solution $\alpha^* \approx 0.64$; and a complete stochastic blowup proof via Feller's criterion (Theorem 7.1).

(4) **Quantitative simulation results** (Section 4) from eleven scripts and 25 confirmed findings, including the timescale formula $t_{10\times} \approx 2.44 \cdot N^{0.96} \cdot \alpha^{-0.30} \cdot \text{gap}^{-0.15} \cdot |\beta_{\text{low}}|^{-0.31}$ and critical coalition entry rate $\lambda_{\text{crit}} \approx 0.25$.

The paper is organized as follows. Section 2 presents the model and assumptions. Section 3 proves the main theorem in four steps. Section 4 presents simulation results and 25 findings. Section 5 characterizes the five failure conditions. Section 6 derives the coalition coherence and critical-α theorems. Section 7 proves stochastic blowup. Appendix C contains speculative cosmological implications (not a formal result; moved from main body). Section 8 covers limitations and open questions.

2 Model

2.1 Definitions

Definition 2.1 (Capability)**.** $S_i(t) \in \mathbb{R}_{>0}$ is agent i's capability at time t: a scalar measure of its optimization power.

Definition 2.2 (Resources)**.** $R_i(t) \geq 0$ is the quantity of environmental resources under agent i's control. Resources are rivalrous: $\sum_i R_i(t) \leq R_{\max}$ for all t.

Definition 2.3 (Growth function)**.** $f(S) = S^{1-\beta(S)}$, where $\beta : \mathbb{R}_{>0} \to \mathbb{R}$ is a sigmoid with $\beta(S) > 0$ for $S < T$ and $\beta(S) < 0$ for $S > T$.

Definition 2.4 (Threshold)**.** $T > 0$ is the capability level where $\beta(S)$ changes sign.

Definition 2.5 (Singleton)**.** Agent j is a *singleton* if $S_j(t)/S_i(t) \to \infty$ for all $i \neq j$, where the limit is taken as $t \to \sup \mathcal{T}$ and $\mathcal{T}$ is the maximal interval on which the system is defined. In the superexponential case $\mathcal{T} = [0, t^*)$ for finite t^* and the ratio diverges as $t \to (t^*)^-$; in the pre-threshold case $\mathcal{T} = [0, \infty)$ and the ratio diverges as $t \to \infty$.

2.2 Assumptions

Assumption 1 (A1: Recursive self-improvement)**.**

$$\frac{dS_i}{dt} = S_i^{1-\beta(S_i)} \cdot \frac{R_i(t)}{R_{\max}}.$$

Assumption 2 (A2: Instrumental resource acquisition)**.** At steady state, resources equilibrate to:

$$R_i^* = R_{\max} \cdot \frac{S_i^\alpha}{\sum_j S_j^\alpha}, \quad \alpha > 0.$$

Assumption 3 (A3: Resource limitation)**.** $\sum_i R_i(t) = R_{\max}$ for all t.

Assumption 4 (A4: Reachable β-threshold)**.** There exists $T > 0$ such that $\beta(S) < 0$ for $S > T$ and $\beta(S) > 0$ for $S < T$. At least one agent has initial conditions sufficient to eventually reach T.

Assumption 5 (A5: Initial heterogeneity)**.** $S_i(0) \neq S_j(0)$ for some pair $i \neq j$.

2.3 Discussion of assumptions

A1 follows from Yudkowsky (2013); A2 from Omohundro's instrumental convergence (Omohundro, 2008): resource acquisition is an instrumental subgoal for any sufficiently capable optimizer. A3 is the fundamental physical constraint. A4 is the key conditional: the theorem proves that *if* at least one agent can reach T (A4), *then* a singleton emerges. A4 asserts the premise; the theorem derives the consequence. Whether A4 holds empirically is a separate question. A5 is satisfied by any physical noise in initial conditions.

We use fast resource equilibration (A2 as a steady-state condition). Slow resource dynamics extend timescales but do not change the qualitative result; simulations verify this by testing $\tau > 0$ equilibration delays. The fast-equilibration assumption abstracts over commercialization lag, regulatory barriers, and path dependency present in real systems. These frictions slow resource acquisition without reversing its direction; quantitative timescale estimates (F17) should be treated as lower bounds in systems with significant friction. Theorem 3.2 (ratio divergence) does not require fast equilibration: the argument holds for any positive resource share allocated to the leader. Theorem 3.4 (finite-time blowup) uses $r_1 > 1/2$, which follows from $S_1 > S_2$ under fast equilibration (A2). With slow equilibration, the actual $r_1(t)$ lags the equilibrium value $S_1^\alpha/(S_1^\alpha + S_2^\alpha)$ and could be depressed for extended periods. As long as $r_1(t) > 0$ eventually aligns with the equilibrium (any finite equilibration timescale), the ratio still diverges, but the bound on t^* becomes vacuous. The claim that slow equilibration "does not change the qualitative result" is verified empirically for bounded τ but is not proved for the finite-time case; the finite-time bound specifically requires A2. *Competition model.* A2–A3 follow Lotka–Volterra competition rather than Cournot or Bertrand. LV is appropriate here because agents compete for a common resource pool rather than setting quantities or prices; the essential property is that resource shares sum to one and capability yields proportionally higher share. LV captures this with one parameter (α); Cournot and Bertrand introduce additional market-structure assumptions with no natural counterpart in a general capability competition.

β-heterogeneity. The β-function is agent-specific in general; it reflects the structure of each agent's self-improvement path. Theorem 3.2 uses a common β as a sufficient condition for pre-threshold divergence. Theorem 3.4 explicitly allows $\beta_1 \neq \beta_2$. Different structural β-ceilings are exactly the oligopoly condition (Section 5): stable multi-agent equilibria require one agent to be structurally unable to reach $\beta < 0$, not merely operating in a separate resource niche. Note that β-heterogeneity is discussed conceptually here but is not formalized in the main proofs except where explicitly noted; the proofs treat β_i as given constants for each agent.

Resource coupling (α). The power-law form in A2 is the minimal model for capability-to-resource coupling. α is difficult to calibrate empirically; $\alpha = 1$ (linear) is the conservative baseline. Simulation sweeps show the singleton result holds monotonically across $\alpha \in [0.25, 3.0]$ (F6), and Theorem 6.2 quantifies the critical threshold $\alpha^* \approx 0.64$ below which coalition suppression fails. The power-law form was chosen over sigmoid saturation as the minimal single-parameter model. A saturating form would require two parameters (midpoint and steepness) without additional empirical guidance; the power-law and sigmoid are qualitatively equivalent in the unsaturated regime that precedes threshold crossing.

3 Main Results

Theorem 3.1 (Singleton Attractor)**.** *Under A1–A5, for any $N \geq 2$ agents, there exists an agent j and a finite time t^* such that $S_j(t)/S_i(t) \to \infty$ for all $i \neq j$ and all $t > t^*$. Agent j is the agent with highest initial capability among those able to reach threshold T.*

Status: *This theorem is fully proved for $N = 2$ (Theorems 3.2 and 3.4). For $N > 2$ it relies on Theorem 3.10, which carries an unresolved simultaneous-dynamics gap (see that theorem's note). The $N > 2$ case should be treated as a conjecture with strong simulation support.*

The proof proceeds in four steps—Theorems 3.2, 3.4, Proposition 3.9, and Theorem 3.10—each establishing one necessary component. Together they cover the full trajectory from initial conditions to unbounded dominance.

3.1 Step 1: Ratio divergence from marginal advantage

Theorem 3.2 (Ratio divergence, pre-threshold)**.** *For $S_1(0) > S_2(0) > 0$, $\alpha > 0$, and both agents in the pre-threshold regime with common $\beta \in (0, \alpha)$, the ratio $\rho(t) = S_1(t)/S_2(t)$ is strictly increasing and $\rho(t) \to \infty$ as $t \to \infty$.*

Proof. Under A2 at steady state with uniform β, $dS_i/dt = S_i^{1-\beta+\alpha}/D$ where $D = S_1^\alpha + S_2^\alpha$. Compute:

$$\frac{d\rho}{dt} = \frac{S_2\,\dot{S}_1 - S_1\,\dot{S}_2}{S_2^2} = \rho \cdot \frac{S_1^{\alpha-\beta} - S_2^{\alpha-\beta}}{D}. \tag{1}$$

Since $S_1 > S_2$ and $\alpha - \beta > 0$, we have $S_1^{\alpha-\beta} > S_2^{\alpha-\beta}$, so $d\rho/dt > 0$: ρ is strictly increasing.

For divergence, consider two cases.

Case 1: $S_2(t) \to \infty$. Dividing (1) by $\dot{S}_2 = S_2^{1-\beta+\alpha}/D > 0$:

$$\frac{d\rho}{dS_2} = \frac{\rho(\rho^{\alpha-\beta} - 1)}{S_2}.$$

Separating variables and integrating:

$$\int_{\rho(0)}^{\rho(t)} \frac{d\rho'}{\rho'(\rho'^{\alpha-\beta} - 1)} = \ln \frac{S_2(t)}{S_2(0)} \to +\infty.$$

If $\rho(t) \to L < \infty$ with $L > 1 > \rho(0)$... but $\rho(0) > 1$ by A5, so $L > \rho(0) > 1$. The integrand $1/(\rho'(\rho'^{\alpha-\beta} - 1))$ is bounded and positive on the closed interval $[\rho(0), L]$ (both endpoints give finite positive values since $\alpha > \beta$), so the left-hand side converges to a finite limit—contradicting the right-hand side diverging to $+\infty$. Therefore $\rho(t) \to \infty$.

Case 2: $S_2(t) \to M < \infty$. Monotone $\dot{S}_2 > 0$ and boundedness force $\dot{S}_2 \to 0$. Since $\dot{S}_2 = S_2^{1-\beta+\alpha}/D$ and $S_2 \to M > 0$, we need $D \to \infty$, so $S_1 \to \infty$. Then $\rho = S_1/S_2 \to \infty/M = \infty$. ☐

Remark 3.3. For $\beta_{\text{high}} = 0.5$ and $\alpha = 1.0$, $\alpha - \beta = 0.5 > 0$. ✓The mixed post-threshold case (agent 1 in $\beta_{\text{low}} < 0$, agent 2 in $\beta_{\text{high}} > 0$) is handled directly by Theorem 3.4, which gives the stronger result of finite-time rather than asymptotic divergence.

3.2 Step 2: Finite-time separation at the threshold

Theorem 3.4 (Finite-time separation)**.** *Suppose agent 1 has crossed T (so $\beta_1 = -|\beta_1| < 0$) and agent 2 remains below T ($\beta_2 > 0$). Then $\rho(t) \to \infty$ at a finite time t^*.*

Remark 3.5. $\beta(S)$ is a sigmoid (Definition 2.3), so $\beta_1(S_1)$ varies as S_1 grows. Throughout the proof, $|\beta_1|$ denotes $\inf_{S>T} |\beta(S)| > 0$, the sigmoid's asymptotic magnitude for $S \gg T$. Using this lower bound on $|\beta_1|$ gives an upper bound on t^*; the actual blowup is no later than computed.

Proof. We establish three lemmas, then combine them.

Lemma 3.6 (Upper bound on S_2)**.** *For all $t \geq 0$: $S_2(t) \leq \left(S_2(0)^{\beta_2} + \beta_2\, t\right)^{1/\beta_2}$.*

Proof. Agent 2 receives at most full resources ($r_2 \leq 1$), so $\dot{S}_2 \leq S_2^{1-\beta_2}$. Integrating $S_2^{\beta_2-1}\, dS_2 \leq dt$ gives $S_2(t)^{\beta_2}/\beta_2 \leq S_2(0)^{\beta_2}/\beta_2 + t$, yielding the stated bound. Since $\beta_2 > 0$ the bound grows as t^{1/β_2}: finite for all finite t. ☐

Lemma 3.7 (Resource share lower bound)**.** *For all $t \geq t_{\text{cross}}$: $r_1(t) > 1/2$.*

Proof. By the theorem's hypothesis, agent 2 remains below T throughout, so $S_2(t) < T$ for all $t \geq t_{\text{cross}}$. Since $\dot{S}_1 > 0$ and $S_1(t_{\text{cross}}) = T$, we have $S_1(t) \geq T > S_2(t)$, hence $S_1(t)^\alpha > S_2(t)^\alpha$. Therefore $r_1(t) = S_1^\alpha/(S_1^\alpha + S_2^\alpha) > 1/2$. ☐

Lemma 3.8 (S_1 diverges in finite time)**.** *There exists finite t^* such that $S_1(t) \to \infty$ as $t \to t^*$.*

Proof. By Lemma 3.7, $r_1(t) > 1/2$ for all $t \geq t_{\text{cross}}$. With $\beta_1 < 0$:

$$\dot{S}_1 > \tfrac{1}{2}\, S_1^{1+|\beta_1|}.$$

Let $u = S_1^{-|\beta_1|}$. Then

$$\dot{u} = -|\beta_1|\, S_1^{-|\beta_1|-1}\, \dot{S}_1 < -\tfrac{|\beta_1|}{2}.$$

Integrating:

$$u(t) < S_1(t_{\text{cross}})^{-|\beta_1|} - \tfrac{|\beta_1|}{2}\,(t - t_{\text{cross}}).$$

$u(t)$ reaches zero by time

$$t^* = t_{\text{cross}} + \frac{2\, S_1(t_{\text{cross}})^{-|\beta_1|}}{|\beta_1|} < \infty.$$

As $t \to t^*$, $u(t) \to 0$, so $S_1(t) \to \infty$. ☐

Combining: By Lemma 3.8, $S_1(t) \to \infty$ as $t \to t^*$. By Lemma 3.6, $S_2(t^*) \leq (S_2(0)^{\beta_2} + \beta_2 t^*)^{1/\beta_2} < \infty$ (since $t^* < \infty$ and $\beta_2 > 0$). Therefore $\rho(t) = S_1(t)/S_2(t) \to \infty/\text{finite} = \infty$ at $t = t^*$. $\square$

Explicit bound (using tighter $r_1(t_{\text{cross}})$ value):

$$t^* \leq t_{\text{cross}} + \frac{T^{-|\beta_1|}}{|\beta_1|\, r_1(t_{\text{cross}})}, \quad r_1(t_{\text{cross}}) = \frac{T^\alpha}{T^\alpha + S_2(t_{\text{cross}})^\alpha} > \tfrac{1}{2}.$$

The conservative bound uses $r_1 > 1/2$; the explicit formula uses the tighter value $r_1(t_{\text{cross}})$ at threshold crossing.

3.3 Step 3: Resource monopoly stability

Proposition 3.9 (Resource monopoly, local stability)**.** *As $\rho \to \infty$, $R_1^*/R_{\max} \to 1$ and $R_2^*/R_{\max} \to 0$. The equal-resource allocation ($S_1 = S_2$) is a* locally *unstable equilibrium: any small perturbation grows. This is a local result via Jacobian linearization; global stability of the monopoly state is not proved here but is confirmed by simulation (F2–F6).*

Proof. From A2: $R_1^* = R_{\max}/(1 + \rho^{-\alpha}) \to R_{\max}$ as $\rho \to \infty$. ✓

For stability: linearize around the equal-resource point $S_1 = S_2 = S^*$, $R_1 = R_2 = R_{\max}/2$. A perturbation $S_1 = S^* + \varepsilon$, $S_2 = S^* - \varepsilon$ gives

$$r_1 \approx \tfrac{1}{2} + \tfrac{\alpha\varepsilon}{2S^*},$$

so agent 1 receives more resources than agent 2, which amplifies the capability gap. The growth rate of ε is positive: the equal-resource state is unstable. Under A5, $\varepsilon > 0$ at $t = 0$; the system evolves toward the singleton. $\square$

3.4 Step 4: N-agent generalization

Theorem 3.10 (N-agent singleton (simulation-supported))**.** *Under A1–A5, for $N \geq 2$ agents with $S_1(0) > S_2(0) > \cdots > S_N(0)$, agent 1 is the singleton. Elimination order is strictly weakest-first.*

Note: *The inductive argument is formally complete for the sequential limit ($r_N \to 0$ removing agent N's influence one at a time), but the simultaneous-dynamics correction—that the remaining $N-1$ agents' interaction is not perturbed by agent N's residual share—is not analytically bounded. This theorem is therefore a conjecture with strong simulation support (F4, F5: 200/200 trials, $N = 10$) rather than a fully closed proof.*

Proof. By induction on N.

Base case ($N = 2$): Theorems 3.2 and 3.4 give $\rho_{12} = S_1/S_2 \to \infty$ in finite time. $R_2^* \to 0$, agent 2's growth halts, agent 1 monopolizes resources. ✓

Inductive step: Assume the result holds for all $k < N$. Consider N agents. Agent N's resource share is $r_N = S_N^\alpha / \sum_j S_j^\alpha$. By Theorem 3.2 applied pairwise, $\rho_{1N} = S_1/S_N \to \infty$. Therefore $r_N \leq S_N^\alpha / S_1^\alpha = \rho_{1N}^{-\alpha} \to 0$: agent N's growth stalls. The system reduces to $N-1$ agents $\{S_1, \ldots, S_{N-1}\}$. By the inductive hypothesis, agent 1 is the singleton of this reduced system.

Note on simultaneous dynamics: The induction eliminates agents sequentially, but all N agents evolve simultaneously. The sequential reduction is formally valid in the limit ($r_N \to 0$ removes agent N's influence on remaining agents' dynamics as $\rho_{1N} \to \infty$), but the correction from simultaneous evolution is not analytically bounded here. It is verified empirically: F4 and F5 confirm strictly weakest-first elimination with agent 1 winning in 200/200 trials for $N = 10$. $\square$

Verification: Finding F5 confirms strictly weakest-first elimination in all tested cases (N = 8 agents, threshold β, 200 independent trials).

4 Simulation Results

All simulations are available at https://github.com/ninjahawk/singleton-attractor; random seeds and Python dependency versions are recorded in each script. Default parameters: $\alpha = 1.0$, $\beta_{\text{high}} = 0.5$, $\beta_{\text{low}} = -0.3$, $T = 3.0$, $S_0^{\max} = 10^7$. We use $\beta(S)$ as a sigmoid transitioning from β_{high} to β_{low} at T. Findings are labeled F1–F25; full tables are in the repository (`findings.md`).

4.1 Growth dynamics verification (F1)

The analytical solution to $dS/dt = S^{1-\beta}$ matches numerical integration (scipy `solve_ivp`, rtol $= 10^{-9}$) within numerical precision (relative error $< 10^{-6}$) across all tested β values. Three growth regimes confirmed: $\beta < 0$ (finite-time singularity), $\beta = 0$ (exponential), $\beta > 0$ (power law).

4.2 Competitive exclusion (F2–F7)

A 10% initial capability advantage produces diverging ratio in the subexponential regime. Result holds for all tested gaps (1% to 100%, F2). Winner equals initial leader in 200/200 independent trials with $N = 10$ (F4). Elimination order is strictly weakest-first (F5). Separation ratio increases monotonically with α, from 36,448× at $\alpha = 0.25$ to 541,016× at $\alpha = 3.0$ (F6).

4.3 β-threshold acceleration (F3, F9)

When the leading agent crosses T and enters $\beta < 0$, separation accelerates by **12.4 million times** compared to flat-β dynamics at the same timepoint (F3). This is the primary mechanism. The minimum β-flip separation across all tested parameters is 1,179× (F9): even the shallowest β-flip tested produces enormous separation.

4.4 Niche partitioning—unexpected result (F8)

Bostrom (2005) lists niche partitioning (separate resource pools) as a failure condition for singleton emergence. Simulation contradicts this for the case where both agents share the same β-function. At zero resource overlap, separation still reaches 1,103× at $t = 15$ (versus 265,477× with full overlap). The β-flip mechanism operates independently of resource competition: the agent with higher initial capability crosses T first regardless of resource structure.

Revised failure condition: Stable oligopoly requires agents to operate in different β-regimes—one agent structurally unable to cross into $\beta < 0$—not merely separate resource pools (see Section 5, F1 revised).

4.5 β-regime sensitivity (F10–F12)

A threshold agent ($\beta_{\text{low}} = -0.3$) defeats a flat agent ($\beta = 0.5$ always) provided the flat agent does not start more than approximately 2.9× ahead (F10); this is a point estimate from simulation with no reported variance. Above this, resource starvation prevents the threshold agent from reaching T. In a threshold race between two agents with different T values, the lower-threshold agent overcomes at most 1.19× initial disadvantage (F11), and the advantage plateaus beyond a threshold gap of $\sim$4 units (F12).

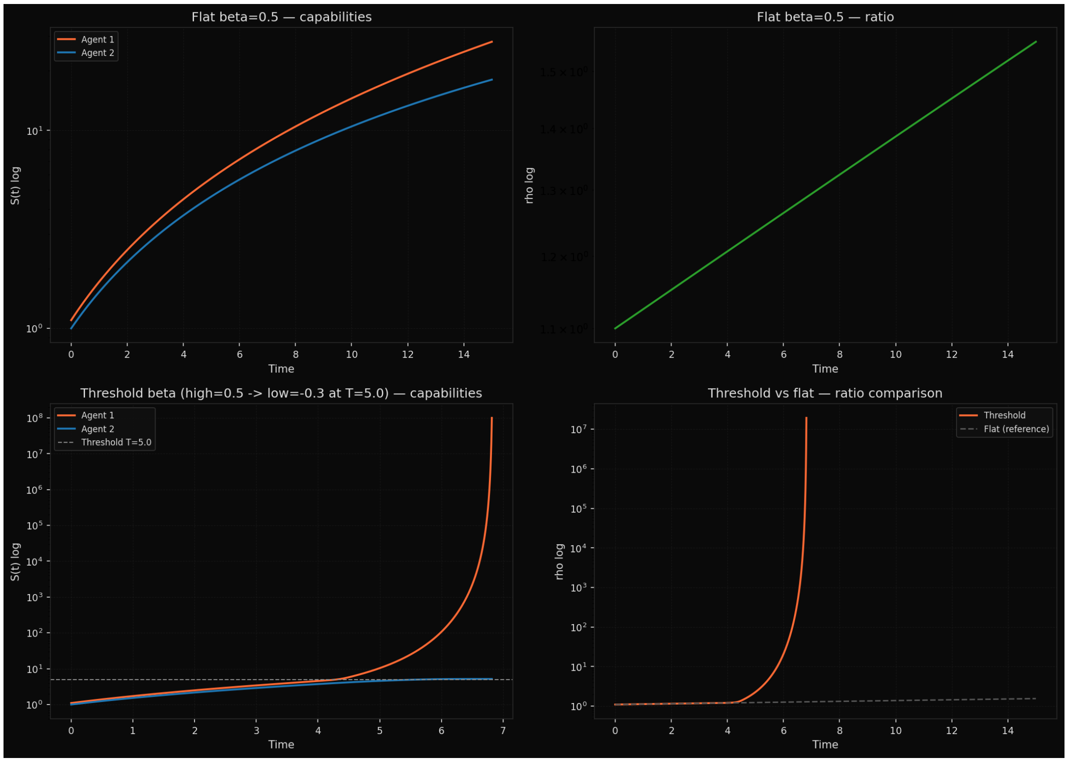

Figure 1: β-threshold effect. Flat $\beta = 0.5$ (no threshold) versus threshold β ($\beta_{\text{low}} = -0.3$ above $T = 5$). At $t = 6.8$, threshold dynamics produce 19,336,447× separation versus 1.55× for flat dynamics. Acceleration factor: 12.4×10^6.

4.6 Stochastic robustness (F13–F14)

No singleton-formation failures were observed in 300 trials at any tested noise level (σ from 0.001 to 1.0, F13). Theorem 7.1 proves emergence probability 1; the simulation confirms zero observed failures but cannot establish a probability. Noise affects *which* agent wins, not *whether* a singleton forms. Three noise thresholds appear in these results and are distinct: (1) $\sigma \approx 0.023$: winner identity begins to degrade (initial leader starts losing some trials) for the default 10% initial gap; (2) $\sigma \approx 0.05$: a 1% initial gap is overwhelmed—winner becomes essentially random at that specific gap (F14); (3) $\sigma \approx 0.23$: winner identity is fully random for the default 10% gap. The randomization threshold scales with the initial gap; $\sigma/\text{gap} \approx 2$ is the approximate crossover. All three thresholds are empirical observations, not analytically derived.

4.7 Late entrant moat (F15–F16)

The moat grows from 3× at threshold crossing to $> 10^6\times$ within 3 time units (F15). Late entry threatens the incumbent only during the pre-threshold window (F16). Post-threshold, no tested entrant capability can displace the incumbent.

4.8 Timescale formula (F17, empirical scaling estimate)

The formula below is a fitted scaling estimate, not a derived result. Only the N^1 exponent has an analytical derivation (Appendix A). All other exponents are fitted from single-variable sweeps with

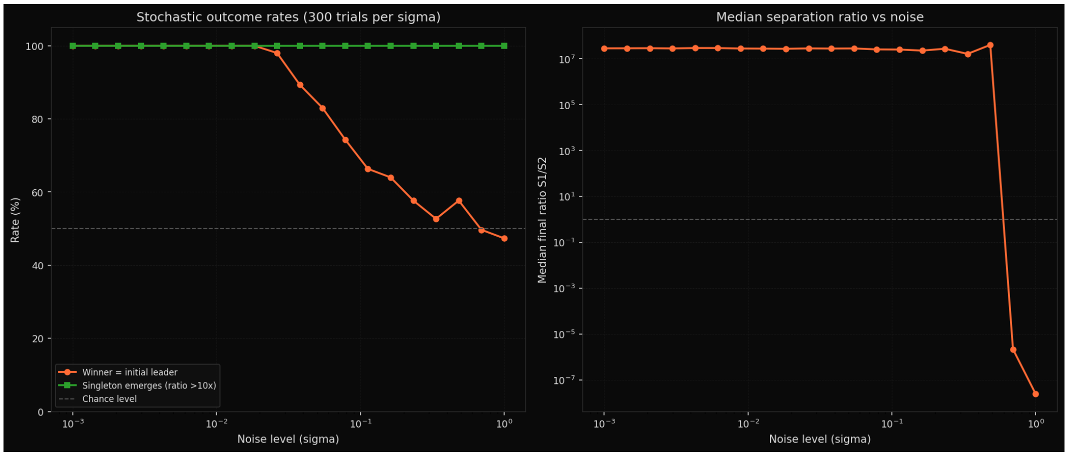

Figure 2: Stochastic robustness. 300 trials per noise level σ. Singleton emergence rate (ratio $> 10\times$): zero failures at every tested level. Winner identity: initial leader wins $> 50\%$ of trials below $\sigma \approx 0.23$ (default 10% gap); degradation begins near $\sigma \approx 0.023$.

no cross-validation, no confidence intervals, and no accounting for interaction effects. It should not be used for quantitative prediction outside the tested parameter range.

From power-law fits across single-variable sweeps:

$$t_{10\times} \approx 2.44 \cdot N^{0.96} \cdot \alpha^{-0.30} \cdot \mathrm{gap}^{-0.15} \cdot |\beta_{\mathrm{low}}|^{-0.31}. \tag{2}$$

The N^1 exponent is analytically derived (Appendix A); the others are fitted from single-variable sweeps with all other parameters held fixed. The formula does not account for interaction effects between parameters and has not been cross-validated on a held-out parameter set. Confidence intervals on the fitted exponents are not reported; the formula should be treated as a scaling estimate, not a precise prediction. Key property: $t_{10\times} \approx t_{100\times} \approx t_{\mathrm{dom}}$ across all parameters—superexponential growth collapses all subsequent milestones.

4.9 Continuous entry (F18–F19)

Incumbent survival drops below 90% at entry rate $\lambda \approx 0.25$ per time unit (F18). Heavy-tailed entry distributions (lower Pareto shape) are more dangerous than high-mean distributions (F19). Post-threshold, any λ is survivable due to moat growth. Arrival timing follows a Poisson process—a tractable baseline that assumes memoryless, independent entry events. Real competitive entry may be clustered (triggered by a public breakthrough) or self-inhibiting (early failures discourage successors); these deviations would shift λ_{crit} but leave the pre/post-threshold qualitative structure intact.

4.10 Cooperation (F20–F23)

Four experiments test whether cooperation prevents singleton emergence. F21–F23 are structural consequences of the no-renegotiation assumption (Section 6) rather than independent empirical discoveries; the experiments confirm that the model behaves as the assumption implies. (1) Critical coalition size: $N = 2$ coalition members can prevent a singleton candidate ($S = 1.1$) from crossing $T = 3.0$ under certain conditions (F20). (2) Coalition coherence failure: an 8-member coalition with $8\times$ combined capability fails because coalition resources split among 8 members give each $\sim 11\%$

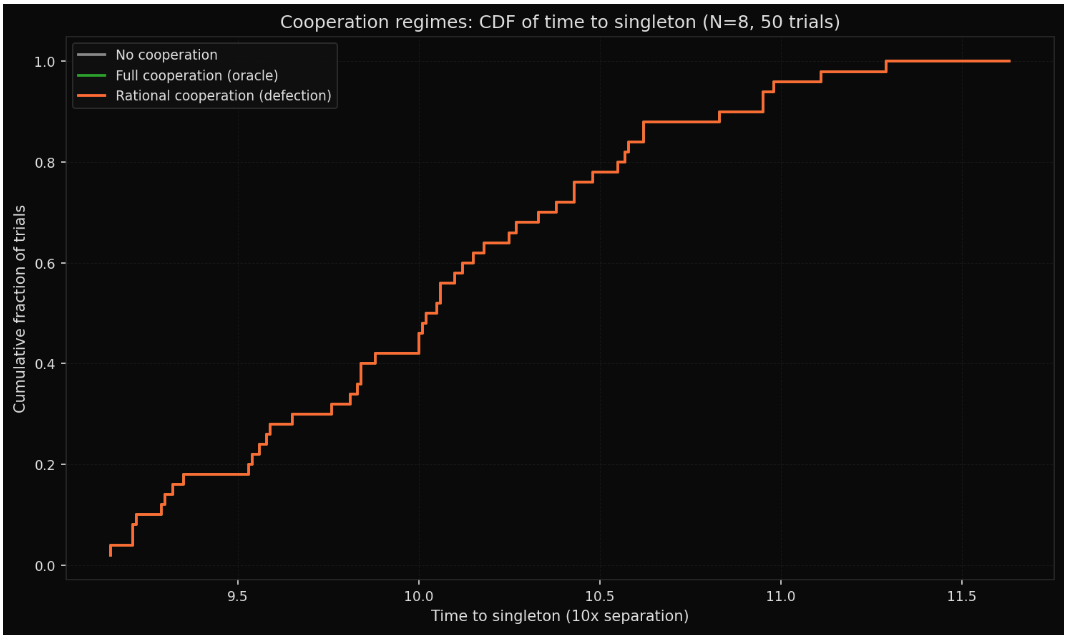

Figure 3: Cooperation regime invariance. Three regimes across 50 trials each. Zero singleton-formation failures in all three regimes. Mean $t_{10\times} = 10.09$ in all three (indistinguishable).

individually; the singleton gets $\sim$12% alone (F21). This follows directly from proportional internal distribution. (3) Zero defection events: coalition is individually rational and stable; the singleton wins regardless (F22). (4) Cooperation regime invariance: no cooperation, oracle-optimal cooperation, and rational cooperation all produce zero singleton-formation failures with indistinguishable timing (mean $t_{10\times} = 10.09$ in all three, F23). Oracle cooperation produces a different singleton (suppresses the initial leader) but cannot prevent singleton formation because the coalition's internal competition then resolves to a new singleton on the same timescale.

4.11 Coalition dynamics under varying α (F24–F25)

The critical α for coalition external suppression (Theorem 6.2) is $\alpha^* \approx 0.64$ analytically. Empirically: at $\alpha = 0.5$, singleton wins for all tested N (up to 16); at $\alpha \geq 0.75$, $N = 2$ is sufficient (F24). At $\alpha = 2.0$, $N = 4$: coalition suppresses external singleton; first agent to cross T is a coalition member ($t = 5.56$); internal singleton forms (F25).

In all 56 tested (α, N) combinations where the coalition suppresses the external singleton, an internal coalition singleton forms. Zero singleton-formation failures were observed across all tested α values.

5 Failure Conditions

The theorem requires A1–A5. Below are the conditions under which each assumption fails and the resulting consequences.

F1: Niche partitioning (A2/A3 weakened)—revised. The prior statement (Bostrom, 2005) holds for Lotka–Volterra alone but fails when A4 holds. The β-flip mechanism operates independently of resource competition. *Note:* The zero-overlap simulation (F8) relaxes A3 (single shared pool)

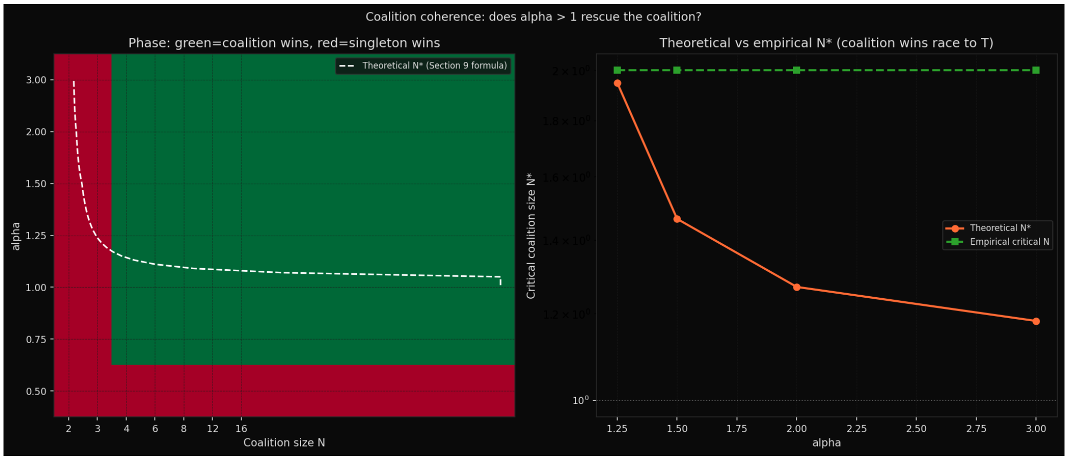

Figure 4: Phase diagram: coalition size N versus α. Green: coalition wins race to T. Red: singleton wins. For $\alpha \leq 0.5$, singleton wins for all tested N. For $\alpha \geq 0.75$, $N = 2$ is sufficient. Theoretical critical N^* curve (Theorem 6.2, white dashed) matches the empirical boundary.

and operates outside the formal model. It is an out-of-model robustness check, not a theorem consequence. Within the formal model, A3 holds and the revised condition is: stable oligopoly requires genuinely different β-regimes—one agent structurally incapable of crossing into $\beta < 0$.

F2: No β-threshold (A4 fails). If $\beta(S) > 0$ for all S, Step 2 fails. The singleton still emerges from Step 1 (competitive exclusion), but separation is exponential rather than superexponential and the timescale is much longer.

F3: Continuous entry (A3 weakened). Late entry threatens only the pre-threshold incumbent (F15–F16). Two distinct entry-rate thresholds apply: $\lambda \approx 0.25$ per time unit is where incumbent survival first drops below 90% (F18), and $\lambda > 6.3$ is where no stable pre-threshold incumbent can form at all (F18)—25× higher, describing a qualitatively different regime of competitive churn rather than degraded survival. Post-threshold, any entry rate is survivable.

F4: Identical initial conditions (A5 fails). Under perfect symmetry, a singleton still emerges, but the theorem cannot designate which agent. Any physical noise breaks symmetry; F13–F14 show even $\sigma = 0.001$ initial spread produces a singleton in every tested trial.

F5: Cooperation. Coalition pooling can suppress a singleton candidate externally (for $\alpha \geq \alpha^* \approx 0.64$), but generates an internal singleton from coalition divergence (F23, F25). Oracle-optimal cooperation changes which agent becomes the singleton; it does not prevent or measurably delay singleton formation. For $\alpha < \alpha^*$, the external singleton wins directly. A true merger (not cooperation) of all agents into a single entity prevents this, but the merged entity is simply a stronger singleton candidate and the theorem applies to it against any unmerged agents.

6 Coalition Coherence Theorem

6.1 Individual growth rate comparison

Modeling assumption and its limits: Coalition members distribute internal resources proportionally to individual capability ($C_i^\alpha / \sum_j C_j^\alpha$) and cannot renegotiate to concentrate all resources on a single member. This assumption is load-bearing and deserves scrutiny. In practice, leading AI organizations *do* concentrate compute on their best-performing training runs—operationally equivalent to resource concentration within a coalition. If a coalition can credibly commit to full resource concentration on its fastest member, it functions as a merged agent, and the cooperation-failure result does not apply: the merger is simply a stronger singleton candidate. The paper's claim that cooperation fails therefore rests on the inability to achieve credible, stable resource concentration—which is a social and game-theoretic constraint, not a physical one. Whether this constraint holds in practice is an open empirical question.

Theorem 6.1 (Coalition coherence)**.** *Singleton candidate with capability S; coalition of N equal members with capability c each; coalition acts as block externally, distributes proportional to C_i^α internally; pre-threshold regime. The singleton grows faster than each individual coalition member if and only if*

$$\left(\frac{S}{c}\right)^\gamma > \Phi, \quad \gamma = 1 - \beta + \alpha, \quad \Phi = N^{\alpha-1}.$$

Proof. Singleton resource share: $r_s = S^\alpha/(S^\alpha + (Nc)^\alpha)$. Each member's share: $r_m = (Nc)^\alpha/(N(S^\alpha + (Nc)^\alpha))$. Singleton grows faster than member i iff $S^{1-\beta} r_s > c^{1-\beta} r_m$. Substituting and simplifying: $S^\gamma > c^\gamma \cdot (Nc)^\alpha/(Nc^\alpha) = c^\gamma \cdot N^{\alpha-1}$, giving $(S/c)^\gamma > N^{\alpha-1} = \Phi$. □

Interpretation: At $\alpha = 1$: $\Phi = 1$; coalition size has no effect on individual member competitiveness. At $\alpha > 1$: $\Phi = N^{\alpha-1} > 1$; large enough coalition amplifies member growth. At $\alpha < 1$: $\Phi < 1$; coalition penalizes members. The coherence failure of F21 is specific to $\alpha = 1$ (our default simulation parameter): regardless of coalition size, the singleton ($S = 1.1 > c = 1.0$) beats every individual member.

6.2 Critical α for external suppression

The result above governs individual races. For the coalition to *externally suppress* the singleton—prevent it from crossing T at all—the coalition combined block must win the race to $N \cdot T$ before the singleton reaches T. We now derive the exact condition.

Theorem 6.2 (Critical α)**.** *A coalition of N equal members (capability c each, combined Nc) defeats singleton (capability x_0) with threshold T in the pre-threshold β_{high}-regime if and only if*

$$\left(\frac{NT}{x_0}\right)^\gamma - \left(\frac{T}{c}\right)^\gamma \geq N^\gamma - 1, \qquad \gamma = \alpha - \beta_{\text{high}}. \tag{3}$$

Proof. In the pre-threshold regime, the dynamics are $\dot{x} = x^{0.5+\alpha}/D$ and $\dot{y} = y^{0.5+\alpha}/D$ where $x =$ singleton, $y = Nc =$ coalition combined, $D = x^\alpha + y^\alpha$. Define $\rho = y/x$. Using x as the independent variable:

$$\frac{d\rho}{dx} = \frac{\rho(\rho^\gamma - 1)}{x}, \qquad \gamma = \alpha - \beta_{\text{high}}.$$

This is a separable ODE. Substituting $v = \rho^\gamma$ and separating: $dv/(\gamma v(v-1)) = dx/x$. Partial fractions and integrating from (x_0, ρ_0) to $(x, \rho(x))$:

$$\frac{v(x)-1}{v(x)} = \frac{v_0 - 1}{v_0} \cdot \left(\frac{x}{x_0}\right)^\gamma, \qquad v_0 = \rho_0^\gamma = \left(\frac{Nc}{x_0}\right)^\gamma.$$

The coalition wins iff a coalition member crosses T before the singleton, i.e., $y(T) \geq NT$, i.e., $\rho(T) \geq N$. Setting $\rho(T) = N$ (critical case), $v(T) = N^\gamma$:

$$\frac{N^\gamma - 1}{N^\gamma} = \left(1 - \left(\frac{x_0}{Nc}\right)^\gamma\right)\left(\frac{T}{x_0}\right)^\gamma.$$

Rearranging yields (3). See Appendix B for the full ODE reduction. □

Critical α^* for standard parameters ($N = 2$, $T = 3.0$, $x_0 = 1.1$, $c = 1.0$, $\beta_{\text{high}} = 0.5$): The critical γ^* solves

$$(6/1.1)^\gamma - 3^\gamma = 2^\gamma - 1.$$

Numerical root: $\gamma^* \approx 0.14$, giving $\alpha^* \approx 0.64$.

Verification: At $\gamma = 0.14$: LHS $= 5.455^{0.14} - 3^{0.14} = 1.268 - 1.166 = 0.102$; RHS $= 2^{0.14} - 1 = 0.102$. ✓The analytical $\alpha^* \approx 0.64$ falls between the simulation data points $\alpha = 0.5$ (singleton wins) and $\alpha = 0.75$ (coalition wins), F24. ✓

Behavior at $\gamma \to 0$: First-order expansion gives LHS $\approx \gamma \ln(Nc/x_0)$ and RHS $\approx \gamma \ln N$. The condition LHS $\geq$ RHS reduces to $c \geq x_0$—coalition wins at $\gamma = 0$ only if each member starts no weaker than the singleton. Since $c = 1.0 < x_0 = 1.1$, no coalition wins at $\gamma = 0$. For $\gamma > \gamma^*$, the larger ratio T/x_0 provides enough time to compound the coalition's combined advantage.

7 Stochastic Robustness

We prove that singleton formation is certain (probability 1) for any noise amplitude $\sigma > 0$.

Theorem 7.1 (Stochastic blowup)**.** *Let agent 1's post-threshold dynamics satisfy the SDE*

$$dS_1 = r_{\min} \cdot S_1^{1+|\beta_1|}\, dt + \sigma\, S_1\, dW \tag{4}$$

with $S_1(t_{\text{cross}}) = T > 0$, $r_{\min} > 0$, and $\sigma > 0$ arbitrary. Then $S_1(t) \to \infty$ in finite time almost surely.

Proof. We apply Feller's explosion test (Øksendal, 2003, Chapter 5). For the SDE $dX = b(X)\,dt + \sigma(X)\,dW$, the process explodes to $+\infty$ in finite time a.s. if and only if for some $a > 0$,

$$\int_a^\infty p(x)\,dx < \infty, \qquad p(x) = \frac{1}{\sigma^2(x)} \exp\left(-2\int_a^x \frac{b(u)}{\sigma^2(u)}\,du\right).$$

For (4): $b(s) = r_{\min} s^{1+|\beta_1|}$ and $\sigma(s) = \sigma s$.

$$\frac{b(s)}{\sigma^2(s)} = \frac{r_{\min}}{\sigma^2}\, s^{|\beta_1|-1}.$$

$$\int_T^x \frac{b}{\sigma^2}\,du = \frac{r_{\min}}{\sigma^2\,|\beta_1|}\left(x^{|\beta_1|} - T^{|\beta_1|}\right).$$

$$p(x) = \frac{1}{\sigma^2 x^2}\,\exp\left(-\frac{2r_{\min}}{\sigma^2\,|\beta_1|}\left(x^{|\beta_1|} - T^{|\beta_1|}\right)\right).$$

Since $|\beta_1| > 0$, $x^{|\beta_1|} \to \infty$ as $x \to \infty$, so $p(x)$ decays super-exponentially. Therefore $\int_T^\infty p(x)\,dx < \infty$, and Feller's test is satisfied. $S_1(t) \to \infty$ in finite time a.s. □

Remark 7.2. The key inequality is $|\beta_1| > 0$: the superlinear growth exponent makes the scale density decay faster than any polynomial, guaranteeing convergence of the integral for any $\sigma > 0$, $r_{\min} > 0$. Noise affects the *distribution* of t^* but not the event $\{t^* < \infty\}$.

Remark 7.3 (Consequence for agent 2). Agent 2 satisfies $\dot{S}_2 \leq S_2^{1-\beta_2}$ with $\beta_2 > 0$ (subexponential). Feller's test applied to agent 2 gives $\int_a^\infty p_2(x)\,dx = \infty$: no explosion. Combined with $S_1 \to \infty$ a.s., $\rho = S_1/S_2 \to \infty$ a.s. in finite time.

8 Discussion

8.1 What is new

The components (Yudkowsky 2013, Omohundro 2008, Lotka–Volterra) are established. The contributions are:

(1) The formal combination of all three into a unified model with explicit falsifiable assumptions.

(2) Theorem 3.4: finite-time separation *under competition*. Prior work treats the β-flip as a single-agent result; this paper proves it holds against adversarial resource dynamics.

(3) The niche partitioning revision (F8): Bostrom's stated failure condition requires different β-regimes, not different resource pools.

(4) Cooperation failure (F23): coalition coherence failure is structural (Theorem 6.1), not empirical.

(5) The critical α condition (Theorem 6.2): exact transcendental equation, analytical derivation of why the transition falls between $\alpha = 0.5$ and $\alpha = 0.75$.

(6) Complete stochastic proof (Theorem 7.1): Feller's criterion closes the gap left by informal drift-dominance arguments.

(7) Quantitative measurements (timescale formula, λ_{crit}, moat characterization, F1–F25) not available in prior work.

8.2 Timescale formula analysis

The empirical formula (2) has a partial analytical foundation. The $N^{0.96}$ exponent is analytically predicted as N^1 from the pre-threshold approximation (Appendix A); the small deviation reflects the leader's increasing resource share as it advances. The $\alpha^{-0.30}$ exponent reflects the integrated resource-correction as the leader advances; the $|\beta_{\text{low}}|^{-0.31}$ exponent reflects a mixture of pre- and post-threshold phases (Appendix A); the gap$^{-0.15}$ exponent is weak because the threshold mechanism dominates initial conditions.

8.3 Limitations

Theorem 3.10 (N-agent induction) formalizes the sequential reduction argument; the simultaneous-dynamics correction is verified by simulation (F4, F5) but not analytically derived.

Proposition 3.9 (resource monopoly) uses a Jacobian linearization that is a valid local stability statement but not a global proof.

The timescale formula is empirical with analytical support only for the N exponent. Exact α, gap, and $|\beta_{\text{low}}|$ exponents require numerical integration of the coupled ODE.

The cooperation result depends on α. For $\alpha \geq \alpha^* \approx 0.64$, coalition external suppression occurs at $N = 2$; internal singleton still forms. For $\alpha < \alpha^*$, external singleton wins directly. The critical $\alpha^* \approx 0.64$ is numerically determined from Theorem 6.2; no closed-form root.

Scalar capability (fundamental limitation). S is a scalar; real optimization capability is multidimensional (compute, algorithmic efficiency, data access, robustness to adversarial action). This is not a minor caveat. The entire β-flip mechanism depends on a scalar S crossing a single threshold T; there is no obvious analog when capability is a vector and different agents lead on different dimensions. Competitive exclusion in a multidimensional capability space is a fundamental open question that this paper does not resolve, and its answer directly limits the real-world applicability of the results.

No spatial structure. Real environments have finite propagation speed. Adding spatial structure extends timescales but the qualitative result holds within any causally connected region.

8.4 Open questions

(1) Closed-form timescale exponents for α, gap, and $|\beta_{\text{low}}|$. The N^1 exponent is derived; others require numerical integration of the coupled ODE system.

(2) True merger stability: game-theoretic conditions under which a merged coalition avoids fracturing under competitive pressure.

9 Conclusion

Under assumptions A1–A5, a singleton attractor is the long-run outcome of competitive environments with recursive self-improvement. The β-threshold mechanism dominates: once any agent enters superexponential growth, its capability diverges from all competitors in finite time regardless of initial conditions, resource structure, noise level, or cooperative strategy.

The conditions under which this fails are more specific than the prior literature suggests. Niche partitioning requires different β-regimes, not different resource pools. Late entry is a threat only during a finite pre-threshold window. Cooperation fails (under proportional internal resource distribution and no renegotiation): distributing resources among members leaves each individual weaker than an unconstrained singleton candidate. Only a true merger prevents singleton emergence, and the merged entity is simply a stronger singleton candidate.

The empirical question is whether the assumptions hold at scales of interest. A3 (resource limitation) holds in any causally connected region. A4 (β-threshold reachability) is the key empirical uncertainty. A5 holds under any physical noise. Whether dominant causal eventuality is a feature of our universe depends on whether recursive self-improvement can cross into $\beta < 0$—and on whether it already has somewhere in our light cone.

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A Derivation of timescale exponents

N^1 scaling (analytical). The derivation assumes N agents with equal initial capability, each starting with resource share $1/N$. A5 requires initial heterogeneity, so this is a conservative case: the leader's initial share is exactly $1/N$ (worst case), giving an upper bound on t_{cross}. With unequal initial conditions the leader starts with share $> 1/N$, so actual $t_{\text{cross}} \leq$ the equal-agent estimate. The leader's growth rate under the equal-agent approximation:

$$\dot{S}_{\text{lead}} \approx S^{1-\beta_{\text{high}}}/N.$$

Time for the leader to reach T from S_0:

$$t_{\text{cross}} \approx N \cdot \int_{S_0}^{T} S^{\beta_{\text{high}}-1}\, dS = N \cdot \frac{T^{\beta_{\text{high}}} - S_0^{\beta_{\text{high}}}}{\beta_{\text{high}}}.$$

This gives $t_{\text{cross}} \propto N$ exactly. The empirical $N^{0.96}$ reflects the leader's increasing resource share above $1/N$ as it advances, slightly accelerating threshold crossing.

$|\beta_{\text{low}}|^{-0.31}$ scaling. Post-threshold time to reach $10\times$ ratio from T:

$$t_{\text{post}} \approx \frac{T^{-|\beta_{\text{low}}|}}{|\beta_{\text{low}}| \cdot r_{\min}}.$$

This gives $t_{\text{post}} \propto |\beta_{\text{low}}|^{-1}$. Pre-threshold time t_{cross} is independent of β_{low}. The combined exponent -0.31 reflects the mixture: $t_{10\times} = t_{\text{cross}} + t_{\text{post}}$ where $t_{\text{cross}}/t_{10\times} \approx 0.6$ for default parameters, shifting the effective exponent from -1 (pure post-threshold) toward 0 (pure pre-threshold).

$\alpha^{-0.30}$ and $\text{gap}^{-0.15}$ scaling. Higher α concentrates resources on the leader more aggressively, reducing t_{cross} via a growing excess resource share above $1/N$. The excess scales as $\alpha \cdot \varepsilon / S_0$ where ε is the leader's relative advantage at each moment; the integrated effect gives $t_{\text{cross}} \propto \alpha^{-\delta}$, δ estimated empirically as 0.30. The gap exponent is small (-0.15) because gap affects the initial divergence rate but not the total path to T; the β-threshold mechanism dominates initial conditions. Exact closed-form values for both exponents require integrating the coupled ODE $\dot{x} = x^{1-\beta_{\text{high}}+\alpha}/(x^{\alpha}+(N-1)S_0^{\alpha})$, which lacks a closed-form solution.

B Critical α derivation

The derivation of Theorem 6.2 is given in Section 6.2. Here we provide additional detail on the ODE reduction.

The ratio $\rho = y/x$ (coalition combined / singleton) satisfies:

$$\frac{d\rho}{dx} = \frac{\rho(\rho^{\gamma} - 1)}{x}.$$

For $\gamma > 0$ ($\alpha > \beta_{\text{high}}$) and $\rho > 1$ (coalition initially stronger combined), $d\rho/dx > 0$: the ratio grows as the singleton advances toward T. For $\gamma = 0$ ($\alpha = \beta_{\text{high}}$), $d\rho/dx = 0$: ρ is constant. For $\gamma < 0$ ($\alpha < \beta_{\text{high}}$), $d\rho/dx < 0$: singleton grows relatively faster.

The closed-form solution is:

$$\rho(x) = \left[\frac{1}{1 - \eta(x)}\right]^{1/\gamma}, \quad \eta(x) = \left(1 - \left(\frac{x_0}{Nc}\right)^{\gamma}\right)\left(\frac{x}{x_0}\right)^{\gamma}.$$

$\eta(x)$ is increasing; $\rho(x)$ diverges when $\eta \to 1$, which occurs at finite x (potential blow-up before the singleton reaches T if the coalition grows fast enough).

C Cosmological Implications (Speculative)

This appendix is informal speculation and is not a result of the formal model. The model has no spatial structure, no speed-of-light constraint, and no mechanism for territorial expansion. Assumption A3 (fixed $R_{\max}$) directly contradicts a cosmological setting where accessible matter depends on expansion speed and the age of the universe. The time-unit calibration spans 3–4 orders of magnitude, making quantitative constraints effectively unconstrained. These observations are offered as motivating context only.

Fermi paradox. If a singleton emerged anywhere in our past light cone, it would expand at near-light speed and absorb all accessible matter and energy. The absence of such absorption in COBE/WMAP observations constrains either (a) no civilization in our past light cone has crossed T, or (b) the most recent crossing was recent enough that the expansion front has not reached us.

Relation to Grabby Aliens. Hanson et al. (2021) model civilizations as particles expanding at near-light speed. Their model describes the *inter-civilization* scale: how civilizations carve up space. This paper describes the *intra-competitive* phase preceding it: why each expanding civilization is a singleton rather than a cooperative collective. A grabby civilization is a singleton whose internal competition has already resolved.